

# ISHA PATRO

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## EDUCATION

### NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING

Master of Science in **Financial Engineering**

GPA: 3.5/4.0

Brooklyn, NY

Expected 05/27

### VELLORE INSTITUTE OF TECHNOLOGY, AP

Bachelor in **Computer Science**

GPA: 3.6/4.0

Andhra Pradesh, India

07/21

## PROGRAMMING & TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, C++, R, VBA

**Web Development:** React, RESTful APIs, AG Grid, Streamlit, Dash, Tablue

**Data Science:** Statistical Modeling, Machine Learning, Deep Learning, LLMs, RAG, AI Agents, Causal Inference

**Databases & Cloud:** SQL, Neo4j, q/kdb+, AWS, MongoDB, Microsoft Excel, Bloomberg Terminal

**Quantitative Finance:** Alpha Mining, Financial Modeling, Risk Management, Fixed Income Analytics, Time Series Forecasting

## EXPERIENCE

### J.P MORGAN CHASE & CO.

01/21 - 08/25

*Software Developer Engineer - II, MBS Division - Fixed Income Pricing Team* (Banglore, India)

- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting \$5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

## RESEARCH / OPEN SOURCE CONTRIBUTION

### FinOKF: Typed Data Format and Certification Protocol for Trustworthy LLM Financial Reasoning

06/25 - Present

- Architecting a typed profile of the Open Knowledge Format with a certification layer whose soundness is mechanically proven in Lean 4, re-executing LLM-generated financial calculations in exact rational arithmetic, then stress-testing the system against 100+ adversarial attacks on real SEC EDGAR filing data.

### Narrative Risk Allocation for Commodity Portfolios

01/26 - Present

- Engineering an end-to-end differentiable deep-learning portfolio allocator that embeds 125K+ financial news articles into volatility-spillover signals across 9 commodity futures, achieving a 1.73 Sharpe ratio over a 16-year walk-forward backtest.

### Implied Prepayment Surfaces via Physics-Informed Neural Networks in Agency MBS Markets

11/25 - 03/26

*Under review at Journal of Computational Finance.*

- Built a PDE-constrained neural network that extracts implied prepayment from agency MBS prices and statistically outperforms the option-adjusted spread benchmark out-of-sample, validated with leakage-safe backtesting and deflated performance statistics.

### qutePandas (Open Source Contribution)

06/25 - Present

- Published a high-performance Pandas-compatible execution layer backed by kdb+/q, achieving order-of-magnitude speedups (10x-100x+) on large-scale analytical workloads while preserving Python-first research ergonomics.

## HONORS / AWARDS / COMPETITIONS

- Awarded the prestigious [Numerix Women in Finance Scholarship](#), a **\$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (**\$1,200**) - across all engineering disciplines, 2017-2018

## ACADEMIC PROJECTS

### VORTEX

05/26

- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.

### Monte Carlo Pricing of European Barrier Options

10/25

- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.

### Advanced NASDAQ Market Prediction System with Deep Learning Optimization

08/25

- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.